99 — The Answer

Edition 6 posed the question: how does Europe become a global player once more in an order where Russia governs through personal authority, America through legal coercion, and China through chemical control? The answer lies in two physical levers that no one today considers in tandem — and which only work together under a different form of governance. A growing carbon axis via Giant Juncao and BiCRS, and a storage axis in the German lignite open-cast mines for the plastic fraction from household waste. Production plus storage, three continents together, one measurable result: a net carbon-negative Europe within ten years.

In the foreword to Edition 6, I wrote that Europe must design a lever that its three rivals cannot replicate. This edition delivers that lever — and it proves to be twofold. Two physical phenomena at which every engineer gazes without ever seeing them as one, because the European governance design has placed them in irreconcilable legal categories. Whoever conceives of them in unison perceives the sole route towards European energy and climate sovereignty that can be realised without war, without colonialism, without a technological race to catch up, and without massive investment in new discoveries. The technology exists. The biomass exists. The storage space exists. Only the institution that links them does not exist — and that is precisely what Nova Democratia designs.

The first lever: growth

Giant Juncao is a C4 grass hybrid of *Pennisetum giganteum* × *P. americanum*, developed in 1986 by professor Lin Zhanxi at the Fujian Agriculture and Forestry University. Under normal field conditions, the crop yields three to four hundred tonnes of green biomass per hectare per year, with proper water management up to four hundred and fifty tonnes, and in an experimental record in Papoea-Nieuw-Guinea, even eight hundred and fifty-three tonnes per hectare. Switchgrass — the American reference for cellulosic ethanol — yields five to thirty-five tonnes. Sugarcane — the Brazilian reference — yields seventy to one hundred and twenty tonnes fresh. Per hectare, Giant Juncao is three to ten times more productive than any crop currently used for industrial ethanol production. C4 photosynthesis in its most extreme form: less water per kilogram of biomass, no pesticides, low fertiliser requirements, perennial (planted once, harvested for years), and growth on saline and alkaline soils where nothing else will thrive.

For Zuid-Amerika and Africa, this is the crop with which BiCRS — biomass carbon removal and storage — and the manufacture of bio-ethanol and bio-methanol can be placed on a completely different cost curve than sugarcane or maize could ever achieve. And this is only the beginning. Giant Juncao was stabilised as a hybrid in two thousand and two. Breeding, selection, and — potentially — targeted genetic modification have scarcely taken place over the past twenty years outside the circle surrounding Lin Zhanxi. Those who join the evolution of this crop now will, within ten years, attain a productivity that renders sugarcane and maize commercially obsolete for industrial fuel. Maize went from four tonnes of grain per hectare in nineteen fifty to eleven tonnes in twenty twenty as soon as serious breeding was initiated. Giant Juncao stands at the threshold of that curve. Europe possesses the chemical faculties and the breeding centres — Wageningen, INRAE, ETH Zürich, KU Leuven — that can accelerate this evolution. Braziliё, Argentiniё, Nigeria, Tanzania, and Kenia have the hectares and the climate. The window of time is approximately three to five years before Chinese parties, who have been actively promoting the hybrid in Papoea-Nieuw-Guinea, Fiji, the Centraal-Afrikaanse Republiek, and Zimbabwe since twenty-ten, have built up an inevitable geographical advantage.

The second lever: storage

The second lever lies on European soil, and today it is being destroyed daily. Household waste incineration in Europe emits approximately fifty million tonnes of CO_2 annually, of which about seventy per cent comes from the plastic fraction. Plastic contains an average of seventy-eight per cent carbon — polyethylene and polypropylene up to eighty-six per cent, polyethylene terephthalate sixty-three per cent, polyvinyl chloride thirty-eight per cent. For every tonne of plastic incinerated, two point five to two point nine tonnes of CO_2 end up in the atmosphere. For every kilowatt-hour of electricity provided by a waste-to-energy plant, it emits two point five to nearly three times as much CO_2 as a coal-fired power station, and four to five times as much as a natural gas plant. These are figures from the American EPA eGRID database and confirmed by GAIA, the international network researching waste incineration. European waste incineration plants are the power generators with the highest hidden costs on the continent, and they are disguised as sustainable energy because they happen to process waste.

The alternative route that no one considers: separating plastic at the source — not for recycling, but for storage — and depositing the material in geological space of which Europe is leaving billions of cubic metres empty at a time. The German lignite open-cast mines Garzweiler, Hambach, Inden, and Welzow-Süd will be closed in twenty thirty. Garzweiler has an operational volume of nearly two billion cubic metres, Hambach over two point five billion. Together with the other three, they offer enough space to store the European plastic stream for several centuries. In a stable anaerobic environment — precisely what a sealed open-cast mine provides — plastic remains stable as carbon storage for hundreds to thousands of years. This is not a theoretical model. It is the logic of nature itself: crude oil resides in geological layers because organic carbon received anaerobic containment. We extract that carbon, turn it into plastic, and return it to the same geological space. That is called circularity — the only circularity that works on a planetary scale.

The synergy

The two levers reinforce one another when they are designed together. Cease plastic incineration: thirty-five megatonnes of CO_2 per year saved. Cease energy-intensive plastic recycling that still leaves seventy per cent residual material: enormous amounts of electricity become available for other purposes. Store the plastic fraction in the emptying open-cast mines: for every tonne of plastic buried, approximately two point eight-nine tonnes of CO_2 equivalent are avoided, sequestered for centuries. Simultaneously, Giant Juncao flourishes on tropical-equatorial fields in Zuid-Amerika and Africa, providing biomass for BiCRS installations that actively remove CO_2 from the atmosphere and convert it into a storable form, and for bio-ethanol and bio-methanol factories that provide high-grade fuel for aviation and heavy transport. The net effect: within ten years, Europe moves from its current position — approximately three point four billion tonnes of CO_2 emissions per year — to net zero, and thereafter to net negative.

Not through solar panels on shivering roofs. Not through wind turbines on coastlines where residents protest. Not through nuclear reactors that might be finished in fifteen years. But by rolling out two proven technologies: a crop that already grows, and a storage method that is already geologically proven. The installations that have yet to be built — BiCRS reactors in Brazilië and Nigeria, plastic sorting and transport lines in Germany and the Netherlands — are industrial tasks of a type that Europe routinely delivered within five years in the twentieth century. Nothing in this design requires a breakthrough in fundamental science. It only requires a breakthrough in governance.

The category error

For here lies the problem that Edition 6 already identified, and which comes to its full gravity in this edition. The current European governance design cannot conceive of these two levers together, nor can it execute them individually. Three EU-richtlijnen — the Landfill Directive from nineteen ninety-nine, the Waste Framework Directive from twenty-eight, and the Renewable Energy Directive RED III from twenty twenty-three — together form a framework under which plastic is by definition waste, storage is by definition prohibited, and incineration by definition counts as sustainable energy. Three directives that are individually defensible and which together force one conclusion: burn the plastic, call it renewable, forget the CO_2. At the same time, the RED III criteria for biofuels, the ILUC directives, and the due diligence regulations are structured such that a Brazilian or Nigerian field of Giant Juncao must first undergo six to eight years of compliance procedures before the first litre of ethanol from such an installation may enter the European market as sustainable. In those same eight years, Chinese consortia build three operational installations in the same countries.

This is not ill will. It is a design flaw. European governance classifies plastic as waste, not as a storage medium. It classifies Juncao as an experimental crop, not as infrastructure. It classifies waste incineration as sustainable, not as a major consumer of carbon. Those who think within these categories — and every European official must, for they form the legal framework within which decisions are made — cannot devise the levers of this edition, let alone execute them. The consensus mode, described in Edition 6 as the form of governance that views higher orders as problems, manifests itself here in full: a crop from China that grows tropically, linked to a storage method in German mines, with financing and design from Europe and execution in Brazilië and Nigeria — this entire image is treated in the current governance climate as reckless, colonial, non-circular, and dangerous to stability. It would affect the existing waste incineration industry. It would render recycling subsidies redundant. It would disrupt existing sustainability certification. That is why it does not happen. That is why it cannot even be proposed out loud without a friendly conversation with the legal department.

The institutional condition

Most importantly in this edition, in line with the most important point of Edition 6: these two levers can only be built if Europe first remodels its governance design. Not afterwards. It is not the case that we first build the installations and then adjust the institution. It is exactly the other way around. As long as the European Commission remains in consensus mode, every industrial climate axis will remain stuck in preliminary studies, be parked in a working group, or be discarded as “no longer a priority” after three terms of commissioners. The institutional remodelling that Nova Democratia proposes — a measured Bondsraad, an independent first-order measurement body, a revised division of roles between orders one to four in which measured facts surpass third-order procedures — is not an alternative alongside these strategic levers. It is the condition for them.

This chapter, and the seven instalments that follow, record the design of those levers in technical detail: how Giant Juncao is cultivated and bred, how BiCRS works chemically, how plastic from household waste is sorted and transported to the open-cast mines, which three pilot installations Europe can start within three years, and — in the final instalment — how a European economic Bondsraad can bring this from paper to steel within a single political cycle of five years. It is an engineering plan for a continent that still has its engineers but has lost its engineering mindset. Rediscover that mindset — and Europe shall once again become a global player. Not through brawn. Not through treaties. By building the two levers that are currently lying unclaimed, and soon will be no longer.